nuity of writing from Old Europe to Ancient Crete. At least in Renfrew's book no arguments are offered which would meet such requirements.

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Indo-European and Nostratic

1. Background

1.1. Despite the practice of historical/comparative linguistics over the course of more than a century, the problem of "distant" or "deep" genetic relationship still generates heated controversy. Thus Greenberg's classification of native American languages (1987), according to which nearly all of them belong to a "macro-family" dubbed Amerind, elicited (among other reviews) a trenchant review article by Campbell (1988), which in turn sparked a retort by Greenberg (1989) and a further treatment of "megalocomparison" by Matisoff (1990). Whatever one thinks of Greenberg's claims, we can be grateful that they led Campbell and Matisoff to write their important assessments of the methodology of establishing distant genetic relationship, to which I will refer at several points below.

My purpose is not to develop this methodology in a theoretical way, but rather to examine another theory of genetic relationship of the kind Campbell refers to as a "lumping hypothesis", namely the so-called Nostratic theory, with particular attention to the handling of Indo-European data. Beyond its obvious theoretical and methodological relevance to the problem of distant genetic relationship, such an examination has other justifications as well. The past twenty years have seen a dramatic intensification of work on Nostratic, especially in the Soviet Union. Moreover, this work is largely unknown to European and American linguists, in part because most of it, until very recently, has been published in Russian. This situation has led, in my view, to an unfortunate lack of communication between (especially) Soviet researchers in Nostratic studies and practitioners of "mainstream" Indo-European linguistics, as carried out in the West. As a result, many Western scholars too readily dismiss an interesting and possibly productive linguistic

enterprise (now being diligently pursued by many talented scholars), while on the other hand, some Soviet linguists mistakenly judge the framework they have erected to have been unsuccessfully assailed (if not unassailable), when in fact it has barely been noticed. Thus V. Dybo (in Illič-Svityč 1984, p. 7–10), and similarly Shevoroshkin and Markey (1986, p. xix ff., 80–86), have convincingly rejected many of the criticisms leveled against Nostratic by Počxua (1974, assessing Georgian material), Andronov (1982, assessing Dravidian material) and Serebrennikov (1982, assessing mostly Uralic and Altaic material). But the more general claim that "Illič-Svityč's reconstructions hold very successfully against the critics" (Kaiser and Shevoroshkin 1985, p. 377) calls for serious qualification, insofar as a detailed assessment on the part of a Western Indo-Europeanist is thus far lacking.

1.2. In its more modern form, Nostratic scholarship has antecedents going back to Franz Bopp's Die Kaukasischen Glieder des indoeuropäischen Sprachstamms (1847).[1] The present essay, however, specifically addresses "the Soviet renaissance in Nostratic studies" (Shevoroshkin and Markey 1986, p. ix), i. e. the "classical Nostratic" theory developed initially by Dolgopol'skij (1964, 1965, 1969, 1972) and Illič-Svityč (1964, 1967 a/b, 1968), codified in the latter's monumental comparative etymological dictionary of Nostratic (1971, 1976, 1984, published posthumously under the editorship of V. A. Dybo), continued by numerous Soviet scholars,[2] and endorsed wholeheartedly by the eminent Soviet Indo-Europeanist V. V. Ivanov (1974, 1979, 1980).[3]

[1] See Illič-Svityč's sketch of the history of Nostratic studies (1971, p. 38–45), or the briefer survey by V. Dybo (1978, p. 400–402).

[2] Recent studies include Terent'ev 1979, Dolgopol'skij 1984, Orel and Stolbova 1988, A. Dybo 1989 a, Pejros 1989, Starostin 1989.

[3] For present purposes, it would be otiose to address in detail the two other major systems of Nostratic research, associated with the work of A. Bomhard (1984, 1987) and V. Shevoroshkin (Kaiser and Shevoroshkin 1985, Shevoroshkin 1988, Shevoroshkin and Markey 1986); both involve fundamental methodological difficulties related to their conception of Indo-European, whereas

I shall consider, then, some of the problems with the handling of IE. material in classical Nostratic work. I begin with the question of simple lexical comparisons in the support of Nostratic etymologies, and proceed to a discussion of morphological comparison; I conclude with certain phonological problems, coming to focus on the IE. laryngeals. In using specific linguistic examples, with fact carefully distinguished from conjecture, I have tried to conform to the kind of "scientifically appropriate" constructive criticism which, according to Dybo (in Illič-Svityč 1984, p. 7), the Nostratic theory welcomes.

2. Lexical Correspondences

2.1. Perhaps the best-known problem besetting the comparison of vocabulary across vast reaches of space and time concerns a range of phenomena variously termed descriptive forms, onomatopoeia, phonaesthetics, or ideophones, among other manifestations of this same linguistic tendency. There is no need to belabor the problem (cf. Campbell 1988, p. 598, 600–1), except to comment briefly on its treatment by Illič-Svityč.

For the most part, Illič-Svityč scrupulously marks such questionable items using the cover-term "descriptive". But problems arise when the term "descriptive" is applied too liberally, as a way of excusing the behavior of forms that fail to match Illič-Svityč's expectations. Thus Nostratic **k/iH/V* "sing" (1971, p. 297, no. 164; the slashes indicate uncertainty as to the reconstruction of the medial sequence) is labeled "descriptive" on the basis of forms like ORuss. *gajati* "caw, croak (of crows)" in Indo-European, together with similar Uralic forms. But from a formal point of view, there is nothing irregular about most of the IE. material: forms like Ved. *gā́yati* "sings", ptcple. *gītá-*, Av. *gāθā-* "hymn" impose both a 'long-diphthong' interpreta-

Illič-Svityč and his followers operate with a "classical" version of reconstructed Indo-European.

tion *geh_x-(*i*-) (or *geh_2-(*i*-), cf. Mayrhofer 1990, p.483), which Illič-Svityč improperly dismisses in favor of *$geih_x$- on the basis of the Uralic forms, as well as a velar *g- (and not the palatal *ĝ- his Uralic reconstruction *$kīγV$- would lead one to expect for Indo-European, cf. 4.1. below). This interpretative permissiveness, one assumes, is not unrelated to the designation "descriptive".

This laxness in the interpretation of "descriptive" material is symptomatic of a more general problem: despite the label "descriptive", most such items are treated much like any other: thus, as in the above example, the designation "descriptive" is essentially a hedge against irregularities shown by some forms; apart from these lapses, such forms contribute to the establishment of the Nostratic lexical stock on an equal footing with other (non-descriptive) material.

It would be absurd to insist that all "descriptive" material be excluded from consideration, since some such material may be linguistically useful (as noted, in defense of Illič-Svityč, by Shevoroshkin and Markey, 1986, p.83 n.25). But the uses to which such material is put must be far more carefully controlled. The systematic investigation of iconic phonetic representation (cf. Ultan 1970, Jakobson and Waugh 1979, p.181–234, p.266–70, among many other studies) has established that languages are immensely rich and endlessly creative in phonaesthetic patterning – a fact which is insufficiently appreciated by Illič-Svityč and his followers, who are thus too reluctant to view many "descriptive" forms as the independent creations they may be, no matter how much they may seem to reflect cognates.

2.2. Descriptive material aside, what semantic and formal criteria should be imposed on other lexical comparisons adduced in support of a distant genetic relationship? For this problem, Campbell (1988, p.595–96) aptly referred to Goddard's important paper on the methodology of proving distant genetic relationship (1975, p.254–55). Goddard reminds us, first, that the formal and semantic criteria used to support an etymology in a *known* linguistic family are relatively strict, and second, that the criteria to be used in support of distant etymologies are essentially the same ones; he concludes: "... if these strict criteria have been found necessary for etymologies within KNOWN linguistic families, it is obvious that much stricter criteria must be applied to word-comparisons between languages whose relationship is in question." But Illič-Svityč's Nostratic etymologies suffer from recurrent problems of both form and meaning.

Gross semantic mismatches are rare; but there are many cases in which the conjectured root meaning and the meaning of the IE. forms show a troubling degree of semantic latitude – enough, at best, to rate this kind of comparison rather low on a scale of usefulness. Thus Illič-Svityč reconstructs (1971, p.299, no.168) a Nostratic root *kojHa in the meanings "fat/greasy" (the Russian gloss *žirnyj* has both senses) and "healthy", on the strength of the IE. root *$g^{u}eih_3$-/*$g^{u}i̯eh_3$- ordinarily assumed to mean "live, be alive" (despite Illič-Svityč's additional gloss "be healthy"), and some Uralic material (traced to a Proto-Uralic etymon *kōja), which I forego listing in detail, and which consistently shows meanings like "grease, greasy, fat". But the semantic basis for this connection is unconvincing. The missing link, apart from Illič-Svityč's phony gloss "be healthy" for the IE. root itself, appears more clearly in the BSl. and Celtic material Illič-Svityč cites (OCS. *žiti* "live", Bulg. *gojá* (caus.) "fatten", Pol. *goić* "heal"; Li. *gýti* "recover, get well", Latv. *dzît* "id."; OCorn. *buit,* Bret. *boed* "food"), and in his comment on these forms: "Traces of the ancient meaning "to be healthy, to be well fed" still preserved in BSl. and Celtic." But this is pure fantasy. A form like Latin *vīctus* "food, victuals", which is transparently based on the participial stem of *vīvō* "I live", shows that the Celtic forms (also based on *to*-participles) reflect nothing semantically primeval; and the BSl. material is also easily derivable from a meaning "live". Note further the archaic adjective with *u*-extension *$g^{u}ih_3$-*u̯*(-)*o*- (Lat. *vīvus,* Skt. *jīvá*-), which in BSl. (OCS. *živъ,* Li. *gývas*) and Celtic (W. *byw*) means "alive" (not *"fattened, well-fed; healthy" etc.), as it does elsewhere. Rather strikingly, Illič-Svityč chose not to adduce his own gloss *žir(nyj)*, which in addition to its meanings "fat" and "grease" in Russian has Slavic cognates meaning "food, feed"

et sim. (e.g. Cz. *žír,* Pol. *żyr/żer*): he was no doubt aware that its etymological connection with the "live" root was uncertain, and remains so to this day. (For speculations beyond those found in Vasmer, see Trubačev 1967, p.56, s.v. *žir.*)

2.3. Another pervasive problem concerns Nostratic roots whose alleged IE. descendants, in contrast to the preceding example, show very marginal types of attestation. Consider, for example, Nostratic **ĆarV* 'hardened crust' (1971, p.205, no.47), established on the basis of the set IE. **sker-* 'crust (of ice, snow), crust/rind (of bread etc.)' ~ Ur. **ćarV* 'film; hard(ened)' ~ Dr. **carV* 'rough' ~ Alt. **čar(V)* '[= IE.]'. The IE. material provided is the following (cf. Pokorny 1959, p.933): Norw./Sw. *skare* 'crust (of snow)', OIc. *skrá* 'piece of dried skin', MHG. *schrāde* 'thin, dry'; Li. *skrę̃sti* (pret. *skrendaũ*), *skrèsti* (pret. *skretaũ*) 'be covered with a dry film/crust', ORuss. *skora* 'hide'.

According to well-known principles, the isolation of this material within Indo-European could be used to argue in favor of its relative archaism. But this is too crude a yardstick: each case must be evaluated within the context of what we otherwise know about Indo-European. Here we find the following frequent combination of features: (1) restriction to Germanic and Balto-Slavic (which show many isoglosses that do not reflect ancient IE. material; see Stang 1972, and more recently e.g. Pijnenburg 1989); (2) forms in both branches with initial sequence **sTR-* (generally diagnostic of late formations, especially given the next feature); (3) extreme variation with regard to enlarged stems (**skr-et-* etc.), not only between BSl. and Gmc., but within each branch; (4) partly questionable semantic variation. (Despite Illič-Svityč's assertion to the contrary, some of the forms, especially those meaning 'crust' and 'hide', are easily attributable to IE. **(s)ker-* 'cut' – with *s*-mobile! cf. Russ. *korá* 'bark, rind, crust', *kórka* 'rind, crust' – which underlies many similar enlarged formations in BSl., Gmc. and elsewhere.) Forms like these, then, cannot be used to reconstruct anything very old in Indo-European, much less a primordial root **sker-* meaning 'crust of ice or bread'.

2.4. The possibility of borrowing and other forms of linguistic diffusion, including *Sprachbund*-type areal convergences, constitutes a central problem for distant (as well as less remote) lexical comparisons. Although the problem is well known, its relative complexity and the extent of the uncertainty it introduces into distant comparisons are perhaps not sufficiently appreciated. I refer, once again, to Campbell's discussion (1988, p.599–600), including the important observation that because of the strongly adaptive tendencies of some languages, loanwords often appear to mock regular sound correspondences.[4] It is now clear, moreover, that some of the conventional wisdom about borrowing is in need of revision, such as the idea that words for body parts tend not to be borrowed.[5]

Still, Ivanov has incisively argued that the "substantially complicated task ... of delimiting ... *Sprachbund*-phenomena from a proto-linguistic heritage ... is, in principle, soluble, because archaic components of a grammatical system, identified by means of internal reconstruction, cannot be considered as grammatical phenomena typical for a *Sprachbund*" (1980, p.3); and similarly: "The larger the body of irrefutable grammatical evidence for the primordial relationship among the Nostratic languages becomes, the more substantial the mutuality of shared lexical phenomena becomes" (1980, p.10). Thus lexical comparison alone, no matter how extensive, is insufficient to demonstrate distant genetic relationship. We must also be able to find, in Ivanov's words, "archaic components of a grammatical system" or "grammatical evidence" which can only be explained by genetic relationship (as opposed to chance, language typology, or diffusion).

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[4] On diffusion and related problems, see also Matisoff 1990, p.108–14.

[5] For a detailed study of an extensive series of such borrowings, see A. Dybo 1989b. Thus, despite the seductive semantic and formal match of a body-part term, a reconstruction like Nostr. **ĶawingV* 'armpit' (Ural. **kajŋa-la* ~ Dr. **kavuñkV* ~ Alt. **kʻawiŋi* 'id.'; Illič-Svityč 1971, p.344, no.220) must be used with caution: Illič-Svityč himself notes that the word is restricted to the Eastern branch of Nostratic, and that this is a rare (indeed, an almost unique) case of a Nostratic trisyllabic form.

3. Morphology

3.1. Unlike Bomhard, who fails to consider grammatical morphemes almost entirely, Illič-Svityč includes a number of such items in his corpus. Despite the serious attention these morphemes receive, there is little sense in most work on Nostratic (apart from the comments by Ivanov just cited) of the fundamental significance of morphological comparisons for the Nostratic hypothesis, or of certain issues affecting the evaluation of morphological comparisons themselves.[6]

3.2. Grammatical morphemes would seem to be exempt from some of the problems affecting lexical morphemes, such as interference from iconic sound symbolism. But morphemes have problems of their own, as well as problems they *do* share with other lexical items (e.g. uncertainties introduced by borrowing or diffusion).

The most basic problem is the relative brevity of most grammatical morphemes (cf. Campbell 1988, p.600, Matisoff 1990, p.111–13). In the absence of reconstructable information about prosodic features like pitch, most alleged Proto-Nostratic grammatical morphemes consist, like those of Indo-European, of only one or two segmental phonemes. Even though the consonant inventory of Proto-Nostratic is extraordinarily rich, the minimal phonetic body of reconstructed Nostratic morphemes opens up vast possibilities for chance correspondences among grammatical morphemes – not only among "Nostratic" languages, but around the world.

This last point hints at a more insidious problem, which in my opinion is insufficiently recognized by researchers in Nostratic. I refer to the fact that some types of *grammatical* morphemes seem to show behavior similar to phonaesthetic patterning. Consider, for example, Illič-Svityč's reconstruction (1976, p.56, no.290; cf. Dolgopol'skij 1964, p.42) of the Nostratic prohibitive particle *mä,* based on the correspondence set Sem.-Ham. **m(j)* [prohib./neg. ptcle.] ~ Kartv. **mā/*mō* [prohib./neg. ptcle.] ~ IE. **mē* [prohib. ptcle.] ~ Dr. **ma-* [prohib./neg. stem] ~ Alt. **(-)mä(-)/*bä-* [prohib./neg. ptcle.]. In connection with these forms, note also Dolgopol'skij's far-reaching claims (1964, cf. Shevoroshkin and Markey's discussion, 1986, p.xvii–xviii) about the relative stability of some fifteen or so types of lexical items, including various negative (and prohibitive) expressions, as well as Ivanov's insistence on the probative value of this particular equation (1977, p.63). But this equation is one of the weakest in the entire corpus, for the simple reason that similar negative expressions can be found almost anywhere – a fact which led Campbell to offer the same example in his discussion of specious resemblances (1988, p.602): he notes the existence of a "negative *ma*" in Afroasiatic, Indo-European (e.g. Skt. *mā́*), Sino-Tibetan, Thai, Algonquian, Quechua and Mayan (not all of which, after all, are "Nostratic" languages, even for the most ardent partisan of the theory). The frequency of nasal consonants in negative morphemes (and similar expressions) was noted already by Otto Jespersen, who did pioneering research on sound symbolism and other types of phonaesthetic patterning.[7] No matter how "stable" some negative morphemes may be in a language over the course of time, such tendencies seriously weaken the value of an etymology like Nostratic **mä.*[8]

3.3. In view of the problems noted in 3.2., it seems clear that not just any morphological comparison will do. For this point, Campbell's discussion of grammatical similarities (1988, p.601 ff.) reminds us of Sapir's conception of a "profound kernel of grammar reflecting genetic origins", a notion developed in more recent Americanist work in terms of so-called "sub-

[6] Thus Dybo (1978, p.410–11) appeals to the existence of varied morphological material in the Nostratic corpus only as an argument against the idea that the similarities shown by the lexical corpus as a whole largely result from ancient contact situations.

[7] See Jespersen 1924, p.314–16 on universal tendencies for initial *m-* in adversative conjunctions (whether or not his fanciful *explanation* for this phenomenon is correct), with reference (316 n.1) to his earlier work on the subject.

[8] The complex evolution of most negative systems, as documented in many classic studies beginning from Jespersen 1917, raises serious doubts as to the supposed "stability" of negative expressions.

merged" or "contextualized" grammatical similarities. Campbell again cites Goddard (1975, p. 259): "It is necessary to show not only that the [grammatical] resemblances are so numerous and detailed as to exclude the possibility of chance as an explanation but also that they are so tightly woven into the basic fabric of the languages that they cannot be explained simply as borrowings." Nevertheless, work on Nostratic shows little appreciation of the importance of this type of arbitrary network of submerged grammatical correspondences. For the IE. material in Illič-Svityč's corpus, morphological comparisons are vitiated by a failure to represent the true complexity of the data; paradoxically, this is what prevents Nostratic scholars from finding convincing cases of "deep" morphological comparisons.

As a typical example, consider Illič-Svityč's etymology for the Nostratic adjective-forming suffix **-lV* (1976, p. 20, no. 253), based on the correspondence set Kartv. **-l-/*-el-/*-il-* [ptcple. suff., deverbal noun suff.], **-il-/*-ol-* [dimin. noun suff.] ~ IE. **-l-* [denom'l/deverbal adj. suff.] ~ Ur. **-la-/*-lä* [adj. suff.; noun/dimin. noun suff.] ~ Alt. **-l* [deverbal noun suff.] (cf. [?] Sem. **-al-/*-ul-* [dimin. suff.]; Dr. [Malto only: *-le,* pret. ptcple. suff.])

The extensive list of IE. material Illič-Svityč adduces includes the following forms (given with Illič-Svityč's grammatical descriptions): Arm. *sirel* 'love, to love' (deverb. n. suff.), Hitt. *attala-* 'father's' (adj. suff.), Lat. *humilis* 'on/of the earth' (adj. suff.), *legulus* 'gatherer' (agent suff.), OHG. *tregil* 'carrier' (agent suff.), OCS. *bylъ* 'was' (ptcple. suff.), Toch. A *-l,* B *-lye* (deverbal adj. suff.). But even a cursory look at such forms forces us to admit, first of all, that the *l*-formations of Indo-European are exceedingly diverse in nature, and that secondly, partly owing to this diversity, their precise history (including their possible relationships with one another) is still quite unclear. Thus the relation, if any, between the participal *l*-formations (e.g. Slavic, Armenian) and other essentially nominal formations like Lat. *humilis* (some of which may be quite old: cf. Gk. χθαμαλός (**-h₂lo-*?), like Lat. *similis*: Gk. ὁμαλός, etc.) is unknown. Compare also Hamp's attempt (1982) to etymologize Gk. φίλος on the basis of the Anatolian pronominal genitive in

-l- (an important *l*-formation not mentioned by Illič-Svityč; cf. Hitt. *apēl* [~ Lyd. *bil*] 'eius, illius', *kēl* 'huius' etc., with possible traces elsewhere in Indo-European; see in general Shields 1983). Hamp's analysis (gen. *φι-λ [~ Lyd. *bil,* Hitt. *apēl*; cf. φίλ-τατος, φίλ-τερος, Lac. φιν] → φίλ-ο-) shows that in theory, such a formation may even underlie other forms that appear to be plain thematic *l*-adjectives, which begins to suggest the real complexity of the problems surrounding the IE. *l*-formations. Without a coherent account of these various (no doubt partly distinct) formations (cf. Solta 1970 for a partial attempt), terms like "denominal or deverbal adjective suffix" have little or no value for extra-IE. comparisons; thus IE. data like these – at least, in the form Illič-Svityč presents them – cannot contribute to a convincing "deep" morphological analysis.

3.4. The most explicit statement known to me regarding the importance of "deep" morphological comparisons among Nostratic languages is due to Ivanov, who discusses two such items, one involving the system of verbal desinences, and another, certain facts of nominal inflection and derivation (1980, p. 4–10). But in both cases, the IE. facts which serve as the basis for comparison with extra-IE. Nostratic data are themselves given speculative and controversial analyses, thereby undermining the probative value of such comparisons. The first example (1980, p. 4–8) concerns purported relationships among the IE. primary and secondary endings, the (singular) endings of the perfect and middle (with a glance at the Hittite *ḫi*-conjugation), and certain pronominal endings alleged to underlie some of these desinences. Even if Ivanov's assumptions about the precise interrelationships among these items were made clear (and they are not), almost nothing in Indo-European could be murkier or more controversial than these supposed primordial inter-desinential correspondences (and the supposed inter-paradigmatic correspondences they may – or may not – imply).

At first glance, the second example (1980, p. 8–10) appears more promising: it is based on the IE. heteroclitic neuters (esp. the "*r/n*-stems"), a category with distinctive and well-established morphological features. But Ivanov describes this cate-

gory as showing "alternation between affixes containing *-*n*-*t*- in the oblique cases vs. affixes without this element in the direct cases" (1980, p.9) – a highly unorthodox account. There is no good evidence for an oblique stem *-*n*-*t*-, as opposed to the *-*n*- usually reconstructed. The dental of the oblique stem of forms like Gk. gen. ἥπατος is usually thought to be secondary (cf. ὀνόματος etc.; similarly even Illič-Svityč 1976, p.79), and the status of the dental element in nominative forms like Skt. *yákṛt* (beside e.g. nom *ásṛk*) is too uncertain to be used as evidence for an oblique *-*n*-*t*-.[9]

The reason for positing an IE. oblique in *-*n*-*t*- soon becomes clear. In seeking to validate Illič-Svityč's comparison between the IE. *-*n*- of heteroclitic neuters and other formants with -*n*- in Kartvelian, Uralic, Dravidian, Altaic and (possibly) "Hamito-Semitic" (1976, p.78–81, no.314), Ivanov adduces (citing Shanmugam 1971, p.245) the evidence of more recent research on Dravidian nouns: Ivanov claims that Illič-Svityč's hypothesis of an "undifferentiated oblique form" *-*n*- in an early stage of Proto-Dravidian (cf. Shevoroshkin and Markey 1986, p.80 n.5) is confirmed by the fact that Proto-Dravidian impersonal nouns (which are functionally equivalent to IE. neuters), as well as those in -*r*-, had an increment in -*tt*- < *-*n̲*-*tt*-. This leads Ivanov to assert: "There can be no doubt that the Common Dravidian nominal type in (*-*r*-/) *-*n̲*-*tt*-, preserved in several Dravidian languages as a protolinguistic reflex, corresponds exactly to the Indo-European heteroclitic type in (*-*r*-/) *-*n*-*t*-, also preserved only in isolated relics in some older dialects, e.g., Lat. *iter, itineris* 'road'". But since an IE. oblique stem in *-*n*-*t*- is a construct of the most dubious sort, the comparison is a tainted one. Ivanov concludes (1980, p.9–10): "This Indo-European

[9] It is not at all certain that the "animatizing" -*nt*- of the type Hitt. *išḫanant-* (beside *ešḫar* 'blood') reflects a formation of IE. date, despite Gamkrelidze and Ivanov 1984, p.303–4 and Lindeman 1986. The formation, in any case, is not restricted to *r/n*-stems, and cannot be used to reconstruct an oblique stem *-*n*(-)*t*- for (*inanimate*) *r/n*-stems. The *t*-extensions of forms like Hitt. *gimmant-* 'winter' and *išpant* 'night', even if attributable to "late Indo-European" (see Oettinger 1982), likewise do not support an archaic oblique stem *-*nt*- for *r/n*-stems.

– Dravidian correlation ... is particularly convincing since, in both Proto-Indo-European and Proto-Dravidian, heteroclisis is independently reconstructed as a highly archaic grammatical feature which can be projected onto the very core of their grammatical systems by means of internal reconstruction. This fact alone excludes any presumption that the typologies considered here could have emerged in both Dravidian and Indo-European as the result of an allogenetic relationship." But the strength of this claim is unwarranted: it is not sufficient to demonstrate that both Proto-Indo-European and Proto-Dravidian made use of vaguely similar types of heteroclitic nouns; we must be able to show that the *details* of each formation require us to assume genetic relationship.

3.5. A Nostratic optative morpheme *-*jE* ("*E*" = an otherwise unspecified front non-high vowel) was identified by Illič-Svityč primarily on the basis of facts from Indo-European and Altaic, as shown by his lemma (1971, p.282, no.149) providing the correspondence set IE. *-*i̯*-, *-*i̯eh$_1$*- [opt.] ~ Alt. *-*jE*- (opt., volunt.]. Most of the problems with this comparison are apparent from the lemma itself. First of all, this reconstruction (like many others) is based on only two branches: without additional argumentation or evidence, this sort of basis is generally considered insufficient.[10]

Let us next consider the IE. and Altaic forms themselves, beginning with some observations about the IE. optative, and restricting the discussion, for now, to the athematic optative. While certain details remain unclear (e.g. the analysis of so-called "full-grade" forms: Ved. *dheyā́m*, Gk. θείην etc.), the basic facts are indisputable. Thus, it is clear from the classical equation of the Greek and Indo-Iranian optatives (together with relic-forms in the Latin subjunctive) that the IE. optative reflected a grammatical category much like the subjunctive, and so could be applied, from the beginning, to all persons. In gen-

[10] Illič-Svityč tentatively added Dravidian, on the strength of an Old Tamil 2 sg. opt. marker; but even Illič-Svityč regarded the isolation of this form within Dravidian to be questionable enough to warrant its exclusion from his lemma.

eral, person-restrictions like that seen in Slavic, where the optative remains as a new second-person imperative, represent later developments from this original situation. Note further that (excluding, again, certain problematic forms) the IE. optative morpheme was characterized by a well-established pattern of full-grade/zero-grade ablaut: in primary athematic formations (including present, perfect, and root-aorist optatives), the root appears in the zero-grade throughout the paradigm, while the optative morpheme shows *e*-grade in the singular and zero-grade in the plural. Like similar patterns elsewhere in the IE. athematic verbal system, such patterning is generally regarded as a very old feature in Indo-European.

We have, then, a specific profile that might serve as the basis for a convincing "deep" morphological comparison, if we can only demonstrate that some of the facts of Altaic reflect at least some of these idiosyncratic features in an arbitrary, "submerged" way. To return to Illič-Svityč's lemma, and specifically to the designation "voluntative": this indicates that in Altaic, the relevant material (e.g. Old Turkish *-jin,* Azerbaidjan *-ajim,* both, incidentally, with a poorly-explained final nasal, Mongolian *-ja/-je*) occurs exclusively as first-person markers, both singular and plural. Now perhaps this person-restriction is secondary, much as in Slavic. Or has Indo-European innovated in extending the category to all persons? In either case, *what is the evidence*? Likewise, perhaps the ablaut system of Indo-European is entirely innovative, from the point of view of Nostratic (cf. 4.1. below), in which case the full-grade/zero-grade variation in Indo-European is secondary, and so would not appear (and it does not) in Altaic. But these uncertainties leave us with a slender basis on which to build a morphological etymology, much less a "deep" one, especially since it involves a formant with limited phonetic body.[11] Indeed, the Nostratic optative etymology is vitiated by purely formal considerations, as is evident, again, from Illič-Svityč's lemma: the IE. morpheme is given two versions, one simply the glide **-i̯-* (clearly needed so as to provide a more direct comparison with Altaic), and the other (**-i̯eh$_1$-*) equivalent to the full-grade variant of the athematic optative morpheme discussed above. These forms raise two serious questions: (1) *What is the evidence for the reconstruction with a plain glide?* (2) *Where did the IE. laryngeal come from in the second form?*

Illič-Svityč's answers appear from his analysis of *thematic* optative forms like Skt. *bháret,* Gk. φέροι, Go. *bairai* (which Illič-Svityč derives from a **bher-o-i̯-t*), and from his comment "**-i̯eh$_1$-/*-ih$_1$- ... по-видимому, сложный формант < **-i̯-eh$_1$-*": it is thus the IE. *thematic* optative that shows a simple glide **-i̯-* (a reconstruction given without further comment or reference), which in turn suggests that the athematic suffix is a "complex morpheme", consisting of this same **-i̯-* extended by an otherwise unidentified enlargement **-eh$_1$-*. (Whether this is the stative morpheme **-eh$_1$-*, or something else, we are not told.) But from the standpoint of Indo-European, these analyses are completely unacceptable.

Even in Illič-Svityč's day, traditional thinking about the IE. thematic optative was based on the fundamental principle *entia non sunt multiplicanda*: the *-i-* of the diphthong that characterized the thematic optative was connected with the "*ī*" variant (in pre-laryngealistic terms) of the athematic optative (i.e. its zero-grade form, in current parlance), rather than posit an entirely different thematic optative marker. But this, in effect, is what Illič-Svityč does, except that he attempts to salvage it by claiming that the athematic marker is an enlarged formation – a claim for which there is not a shred of evidence (outside of Altaic), beyond his questionable analysis of the thematic optative itself.

[11] Serebrennikov's argumentation against a Nostratic optative (1982, p.73–74) on the basis of his (mistaken) view of the date of Proto-Nostratic ("40 thousand years ago") is indeed "fatuous" (as Shevoroshkin and Markey observe, 1986, p.82 n.14; similarly Dybo, in Illič-Svityč 1984, p.10, n.10), and his reference to an IE. thematic optative stem **-i̯e-* is hopelessly confused. But his idea that the Turkish optative marker *-aj-* is genetically identical to (and developed out of) a diminutive suffix *-aj-* is a more substantive claim, on which Shevoroshkin and Markey fail to comment; Dybo (in Illič-Svityč 1984, p.10) rejects it out of hand, without explanation.

At present, the most widely accepted view about the thematic optative corresponds to the traditional one: the thematic vowel was followed by the zero-grade form of the optative suffix (thus "*-oi-*" < **-o-ih$_1$-* [traditional "*-o-ī-*"]). This entails, to be sure, the difficulty of explaining why the laryngeal underwent loss and/or assimilation in contexts where we might expect it to remain (or, as in Greek, to be vocalized).[12] According to another approach (Watkins 1969, p.226f.), the thematic optative *does* have a different morpheme from the athematic optative, in view of Indic formations showing a diphthongal element (as in the thematic optative) in roots that otherwise make athematic root optatives (of the type Ved. opt. *gaméma* [~ *gamyā́ḥ*], etc.). In this case, the *-o-* is not the thematic vowel at all, but part of a morpheme **-oi-*. To the extent that these Indic formations can be explained as secondary (cf. Mayrhofer, loc.cit.), it would be preferable not to set up two different athematic optative formations for Indo-European. But even if Watkins' analysis is correct, his morpheme **-oi-* would still not support Illič-Svityč's reconstruction **-i̯-*.

On balance, then, Indo-European provides evidence only for an optative morpheme like the one we reconstruct for athematic formations, with a final (unsegmentable) laryngeal. Nevertheless, if we believe the comparison with Altaic, we are forced to claim that the IE. laryngeal has no Nostratic source, which in turn leads to Illič-Svityč's non-explanation via enlargement. In fact, according to Nostratic theory, the IE. laryngeals have very definite sources (i.e. from Nostratic laryngeals, post-velars and pharyngeals, in both stop and fricative versions; cf. the tables of phonetic correspondences in Illič-Svityč 1971, p.147ff., esp. 149 and 151–52, and similarly Shevoroshkin 1988, p.539: "IE laryngeals ... are reflexes of Nostr. obstruents"). But in this case (as in others), alleged Nostratic equations seem to show IE. laryngeals that have arisen out of nowhere, which is the problem to be addressed in the next section.

[12] For further discussion, see e.g. Mayrhofer 1986, p.131, or Rix's accounts of the Greek developments, 1976, p.74, 231–33, 261–63.

4. Phonology

4.1. I begin by circumscribing some areas of Nostratic phonology (as they intersect with Indo-European) that it seems unproductive to address at the present time.

For the vowels, the best-known topic arises from Illič-Svityč's explanation of the three-way division of IE. gutturals into palatals, velars, and labiovelars (1964, 1968, p.325, cf. also the extended treatment by V. Dybo (1978, p.403–8); more briefly Kaiser and Shevoroshkin 1985, p.398). Illič-Svityč's Nostratic reconstructions seemed to show that IE. labiovelars result from Nostratic sequences with velar or post-velar followed by a rounded vowel (cf. Nostr. **kojHa* ~ IE. **g^{u}i̯eh$_3$-*, 2.2.), IE. pure velars from Nostratic sequences with a following *a* (cf. Nostr. **ćarV* ~ IE. **sker-*, 2.3.), and IE. palatals from Nostratic sequences with velar or post-velar followed by a front vowel (e.g. Nostr. **kENV* ~ IE. **ĝenh$_3$-/*ĝneh$_3$-* 'know' [Illič-Svityč 1971, p.296, no.163]).[13] According to classical Nostratic theory, this implies (*inter alia*) that virtually the entire system of Nostratic vowels was eliminated in Indo-European (although it survives on the level of traces in the quality of the gutturals), and that Indo-European therefore developed in some secondary way its characteristic system of basic *e*-vocalism *cum* complex apophonic alternations. Needless to say, there is considerable uncertainty as to the details of how this transpired, and so it does not seem worthwhile to enter into problems of Nostratic and IE. vocalism for the purposes of this discussion.

Since most Soviet work on Nostratic is based on the framework erected by Illič-Svityč in the mid-sixties, IE. material continues to be cited in its traditional, "Pokornian" shape (with the exception of laryngeals, more on which below) – in other words, *not* using the Gamkrelidze/Ivanov/Hopper reanalysis of IE. consonantism. In contrast, the "new" IE. consonant system forms the basis of Bomhard's work, and Shevoroshkin operates with his own somewhat eclectic version (cf. Shevoroshkin and

[13] Essentially the same process allegedly accounts for the phonetic nature of the three IE. laryngeals, although this part of the theory is disputed by Kaiser and Shevoroshkin, 1985, p.399ff.

Markey 1986, p. xxiv ff.). Once again, the ongoing disagreement (even among Indo-Europeanists) would render any discussion of problems involving the Nostratic and IE. stop sytems excessively complex and abstruse.

The immensely complicated systems of Nostratic sibilants and affricates are mostly based, of course, on non-IE. material. Despite some intensive work in this area (cf. Dolgopol'skij 1974), there is dissatisfaction even among Nostratic researchers about the accuracy of the entire set of correspondences, as recently expressed by Starostin (1989, p. 108). By contrast, the relatively simple system of liquids, nasals and glides is also relatively uninteresting (at least for Indo-European), or, with respect to some points, based on insufficient material.[14]

These considerations leave us with the IE. laryngeals and their putative Nostratic sources. Illič-Svityč and his followers operate with a standard version of Kuryłowiczian (i.e. roughly mid-sixties) IE. laryngeal theory, and seem to be in general agreement about the development of the laryngeals from the point of view of Nostratic. Moreover, the last few years have seen the publication of three major treatments of IE. laryngeal theory (Mayrhofer 1986, Lindeman 1987, Bammesberger 1988) which reveal substantial areas of agreement among Indo-Europeanists on many points (notwithstanding certain persistent substantive disagreements, one of which will be mentioned shortly). In view of the fact that some of the most dramatic advances in our understanding of both IE. phonology and morphology are directly associated with laryngeals, it is incumbent on researchers in Nostratic to cast a more serious glance in the direction of IE. laryngeal theory.

[14] Thus Ivanov (1977, p. 59) points to the interesting correspondence Nostratic (palatal) **ń-* (followed by back vowel) ~ IE. **i̯-* (e.g. Nostr. **ńowda* 'move quickly' ~ IE. **i̯eudh-* 'id.' [Illič-Svityč 1976, p. 90, no. 327]). But of the eight Nostratic etyma with **ń-* plus back vowel (Illič-Svityč 1976, nos. 318–323, 326–27), only three (318, 319, 327) have IE. comparanda. In the unique medial example (Nostr. **/pʲeHńa* 'pasture, protect, care for' [Illič-Svityč 1984, p. 106, no. 373]: IE. **pō-/*pī-* (< **pōi̯-* < **pe/h₁/i-* [sic!]) ~ Ural. **pīńa* (< **pēńa*?) ~ Dr. **pēṇV-*), the reconstruction with **-h₁-*, although required by the Uralic and Dravidian vocalism, is excluded for Indo-European (whence the slant brackets in Illič-Svityč's impossible IE. form **pe/h₁/i-*).

4.2. *Laryngales indogermanicae ex nihilo non oriuntur.* ("Lex Vine") I now examine several types of cases in which Nostratic reconstructions fail to account for well-established IE. laryngeals.

4.2.1. The most pervasive type of instance is also the most difficult to treat, i.e. situations in which a Nostratic root is reconstructed without a laryngeal, but the IE. data point to a *root-final* laryngeal, or at least an archaic variant with final laryngeal, of the type Nostr. **moLV* 'break into pieces' ~ IE. **mel-* 'grind' (Illič-Svityč 1976, p. 69, no. 302): but a **melh₁-* is at least as basic, cf. Ved. *mṛṇáti,* Hitt. *mallai-,* Li. *málti,* Myc. *me-re-ti-ri-ja* (whence, together with Semitic evidence in favor of a root-final laryngeal, Shevoroshkin's "West-Nostr. variant" **maLHV,* with an "unstable laryngeal" lost in East-Nostratic [1988, p. 538]; cf. also Nostr. **kENV* ~ IE. **ĝenh₃-/*ĝneh₃-* 'know', 4.1.). The problem, of course, is bound up with the vexed question of root-enlargements (including laryngeal enlargements) in Indo-European. Thus for virtually all cases of the type Nostr. *TVRV* ~ IE. *TERH-* (to take the most common root-shape for this problem), Illič-Svityč claims that on the basis of the (non-IE.) Nostratic evidence, the root-final laryngeal in Indo-European is "suffixal" (i.e. an enlargement), even when there are no IE. variants of the type *TER-(T-)* ~ *TERH-* for a given root.[15] From the standpoint of Indo-European, this is methodologically illegitimate: we can posit a laryngeal enlargement, *sensu stricto,* only when we have evidence (as often) of a parallel form either with no enlargement or with a different consonantal enlargement.[16] But beyond pointing out this methodological impasse, there is little we can do but look for a different type of case where an enlargement-argument is excluded.

[15] Cf. Illič-Svityč's comment (1971, p. 297) on IE. **ĝenh₃-/*ĝneh₃-* 'know' vis-à-vis Nostr. **kenV* (Kartv. **gen-* etc.): "In IE the stem is extended by a suffixal **-h₃*." Shevoroshkin, again, operates with a Nostratic proto-form **keNHV,* with "unstable laryngeal" (1988, p. 543).

[16] A legitimate example might be Nostr. **mEwV* 'water, moisture' ~ IE. **meu-* 'moist etc.' (Illič-Svityč 1976, p. 62, no. 298, noting enlarged variants **meu-H-* [e.g. Skt. *mū́tra-*], **meu-d-* [e.g. Gk. μύδος], etc.).

4.2.2. We can find such a case if we consider initial pre-consonantal laryngeals. We immediately face the controversial matter of the so-called prothetic vowels of Greek and Armenian. But to the extent that there is a majority position, it is that found in Mayrhofer 1986 and in many of the papers in Bammesberger 1988 (e.g. Beekes, Ruijgh; contra: Lindeman 1987, p.75–86), and it is this position which I now confront with several standard Nostratic reconstructions.

In IE. *h_2ster- 'star', the initial laryngeal is preserved intact in Anatolian (Hitt. *ḫa-aš-te-er-za*); in contrast, we find vocalization of the laryngeal in Greek (ἀστήϱ) and Armenian (*astł*), and loss everywhere else (Lat. *stella* etc.). In other cases we are not so fortunate, but we still have good evidence for the initial laryngeal. Thus in the word for 'man, hero', the initial vowels of Gk. ἀνήϱ, Arm. *ayr,* Phryg. αναϱ are again best taken to represent a vocalized *h_2- (thus IE. *h_2ner-; similarly Gamkrelidze and Ivanov 1984, p.271–72 **Hner*-), even though the word is not attested in Hittite with initial *ḫ*-, and despite the absence of an initial segment elsewhere (Ved. *nár*- etc., cf. Lat. *stella* etc.). This analysis is supported by independent evidence, namely (1) the lengthening of the final vowel of the first member of IIr. compounds like Ved. *sūnára*- 'handsome, vigorous' (cf. *su*- 'good') (Mayrhofer 1986, p.134, and in Bammesberger 1988, p.331), and (2) the *absence* of laryngeal reflex in forms like Hitt. *innaru*- 'strong, manly' (**en-$h_2$nor-u*- 'having manliness inside'), Gk. νωϱεῖ· ἐνεϱγεῖ Hsch. (**h_2nōr-éi̯e/o*-), which probably results from the loss of laryngeal (for roots with liquid) in the vicinity of *o*-vocalism ("de Saussure's/Hirt's Law", cf. Eichner in Bammesberger 1988, p.140–41).

Consider now Illič-Svityč's reconstruction of Nostratic **NajRV* 'man, male' (1976, p.92, no.331), based on the correspondence set IE. **ner*- 'man, male, (manly) strength' ~ Alt. [**ṇi̯arV* 'man, person'] (cf. [?] Ham.-Sem.: OEg. *nr* 'people', *nrj* 'be strong', Avia (Central Cush.) *nierā* 'husband').[17] The loss of

[17] The brackets around the Altaic form indicate that the Proto-Altaic "reconstruction" is based on only a single branch of the family, in this case Tunguz languages, although the documentation there is extensive. (Illič-Svityč also adds Korean *nä* 'man, person' with a query, given the "unclear loss of **r*".) On the status of "Proto-Altaic", see also Kaiser and Shevoroshkin 1985, p.383, n.9.

the Nostratic glide before the **r*- is handled, in classical Nostratic theory, by a regular Nostratic-to-Indo-European reduction process affecting sequences with *y*-glide followed by any liquid or nasal: thus Nostr. **Vjr*- > IE. **Vr*-, Nostr. **Vjm*- > IE. **Vm*-, etc. (cf. Illič-Svityč 1967, p.325–26). But there is no theoretical apparatus (e.g. laryngeal prefixes, root-initial (laryngeal) enlargements, excrescent initial laryngeals, etc.) that can account for the fact that there is no sign in (non-IE.) Nostratic of the initial laryngeal of the IE. form. Such a form, then, presents a serious problem not so easily explained away.

The point was not lost on Illič-Svityč; at the end of his listing of the IE. forms, he observed: "The Greek, Armenian and Anatolian facts can be explained without positing an initial laryngeal." Indeed, they *can* be so explained; but the cost is prohibitive. Apart from the implausible analogies required to explain the initial vowels of forms like ἀνήϱ, note also the consequences of a non-laryngeal interpretation for forms like Ved. *sūnára*-: we would have to assume that the first member of the compound shows an unpredictable length alternation (*su-/sū*-), which we would be forced to project back to Indo-European. Not surprisingly, this is precisely what Illič-Svityč does. Again, the initial vowel of Greek εὐ- 'good' (vs. Ved. *su*-, OIr. *su-/so*- etc.) points to an IE. *h_1su-, supported indirectly by Gk. ὑγιής 'healthy' (< *h_1su-$g^{u}ih_3$-*és*) which shows the otherwise documented laryngeal loss found in the first member of end-stressed compounds (Peters 1980, p.208, n.160, Mayrhofer 1986, p.125, 140).[18] Nevertheless, Nostratic **šuwV* 'good' (Illič-Svityč 1967, p.371) lacks an initial laryngeal, of which there is no trace in Uralic, Kartvelian, Semitic, or – in Illič-Svityč's view – Indo-

[18] Even if Gk. στεϱοπή 'flash (of lightning etc.)' is non-IE. and thus not an example of this phenomenon (Beekes 1987 and in Bammesberger 1988, p.71), the reduction process itself need not be doubted; cf. (despite Beekes, in Bammesberger 1988, p.66) Ved. (TS, Gṛhya-Sūtras; see Mayrhofer 1990, p.510, s.v.) *gru-muṣṭí*- 'handful', with first member *$g^{u}ṛ[h_2]u$-.

European: IE. **sŭ-* 'good' ~ Ur. **šüwa* 'id.' ~ Kartv. **šu-* 'adorn, be pretty' ~ Sem.-Ham. **šw-* 'similar; suitable'.

4.2.3. The problem affects medial position as well, as shown by the IE. word for 'name'. Here, in addition to the Greek and Armenian forms, it is the morphological structure of the word that imposes the laryngeal interpretation: most of the forms are easily explained on the basis of an ordinary protero-kinetic pattern IE. **h₁néh₃-mn̥*, obl. **h₁n̥h₃-mén-* (see Jasanoff in Bammesberger 1988, p.227–28). Other possible approaches, which would have to operate with long and short *o*-grades, have little morphological plausibility, not to mention the difficulty of generating the initial vowels of Greek and Armenian. But Illič-Svityč's Nostratic **nimi* 'name' (1976, p.82, no.317) shows no trace either of the initial laryngeal or of the medial one: Ur. **nime* 'name' ~ IE. **nŏm-* 'id.' (cf. [?] Sem. **nb* (dissim. < **nm*): Ar. *nbʿ* 'announce', Akk. *nbʿ* 'call (by name)', etc.). Starostin (1989, p.119, no.130) provides a slightly different reconstruction **nojmV*, adding comparanda from Altaic (**n̥iomV-* 'name; spell, tale') and Sino-Tibetan (**maiŋ* 'name'). The form **nojmV* allows us to see more clearly the operation of the glide-loss rule (cf. 4.2.2.) that would account, at least, for an IE. "**nŏm-*" (as opposed to Illič-Svityč's **nimi*, which, via a stem-form **nejm-*, would predict an IE. root **nem-* with ordinary *e*-grade); but we are left, as before, with no accounting either for the initial laryngeal, or for the morphological facts of Indo-European that lead us to reconstruct a medial one.[19]

4.3. I conclude with what seems to me an instructive example of a form *without* initial laryngeal, namely IE. **mēms-* (also *mēms-o-*, **mēms-ro-*) 'meat, flesh' (Ved. *mā́ṃ*, *māṁsá-*, OCS. *męso*, Lat. *membrum*, etc.; for initial **m-*, as opposed to **Hm-*: Gk. μηρός, Arm. *mis*). This word, to be sure, has a rather unusual root-shape (including the consistent lengthened grade):

[19] Ivanov (1977, p.58) accepts the Nostratic etymology, and posits an IE. **n(ŏ)m-*. A more explicit acknowledgment of the morphological difficulty appears in Gamkrelidze and Ivanov 1984, p.181, where the IE. form is given an idiosyncratic reconstruction **en(o)mn̥* (with a "basic" *-o-*), and said to be "structurally not entirely clear".

precisely the sort of case we might expect to be illuminated by the extra-IE. Nostratic data. Yet the Nostratic comparanda (mostly from Uralic, with questionable material from Dravidian and Semitic) require positing an initial laryngeal (whence the Nostratic proto-form **Homsa*), which Illič-Svityč then projects onto the IE. form (1971, p.252, no.14): IE. **(H)mēms-* ~ Ur. **omśa* (~ [?] Dr. **ūñc-*) 'meat'.[20] This in turn leads Illič-Svityč to speculate that the IE. form results from the following contamination: "state I" **h₃ems-* X "state II" **h₃mes-* > *(h₃)mems-*. The weakness of this explanation requires no comment, and the arbitrary nature of the methodology should be clear. If the Nostratic form has no initial pre-consonantal laryngeal (or other non-syllabic source for an IE. laryngeal), then IE. data pointing to a laryngeal should be explained away; if, on the other hand, the Nostratic form *does* have such a segment, then we must conjure one up for Indo-European. But this is not the way to practice historical reconstruction, distant or otherwise; and despite the somewhat abstract character of the IE. laryngeals, we know enough about them to see that they cannot be treated in this cavalier way.

5. Concluding Remarks

The extraordinary fascination the Nostratic theory inspires is not to be denied. Ivanov has written (1974, p.54) that "[u]pon reading Illič-Svityč's work, one is left with a sensation ... of exceptional aesthetic perfection", and according to Shevoroshkin (1988, p.538) "it is no exaggeration at all to say that Nostratic reconstructions are much more precise than IE. reconstructions". In many ways, the Nostratic theory is indeed an elegant and well-articulated structure. But with respect to the treatment of Indo-European data, it remains as yet a house of cards.

Calvert Watkins has recently discussed certain new tech-

[20] More questionable "Hamito-Semitic" data are appended to Illič-Svityč's discussion, but excluded from his lemma. See now also Starostin (1989, p.114, no.33) for "Sino-Caucasian" material and Pejros (1989, p.126, no.34) for Austronesian material.

niques of comparative linguistic methodology in the areas of poetics and ethnosemantics – techniques which, according to Watkins, do not extend to what he calls the "Nostratosphere" (1989, p. 798). But his more general message, addressed to the field of linguistics as a whole, is equally applicable to Nostratic: "Real progress in linguistics over the next quarter- or half-century", writes Watkins, "is going to come from precise and profound examination of the languages". The key words are "precise" and "profound". With renewed attention to the kind of precision and depth advocated above, Nostratic studies may yet fulfill their enormous potential to contribute to our understanding of human linguistic history.[21]

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